

ANALOGUE ELECTRONICS IV

EEE 434

LECTURE NOTES ON
(Principles, characteristics and application of feedback amplifiers)

2005

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Pre-lecture Interactive / Self evaluation.

Explain the following terminology:

- feedback in amplifiers (athletic/manual)
- frequency response of an amplifier
- bandwidth of an amplifier
- decibel
- distortion in amplifiers (non linear, frequency and phase (delay))

- Interactive discussion will carry 20% of the total exam
 - Applied knowledge will carry 40%
 - while lecture notes will carry 70%
- Total 100%

The course comprises 4 topics as below:

1. Feedback amplifiers with emphasis on negative type feedback
2. Oscillators
3. Multivibrators/Pulse oscillators
4. Regulated power supplies

THE CONCEPT OF FEEDBACK IN AMPLIFIERS

Definition: Feedback is a process whereby the output of an amplifier is sampled and fed back to the input for necessary correction action. Feedback could either be negative or positive (degenerative or regenerative). In negative feedback the sampled output is now subtracted from the input before being amplified while reverse is the case in positive feedback. Positive feedback tends to make the amplifier unstable and hence breaks into oscillation which is useless as an amplifier. But this phenomenon is being harnessed as useful in oscillators.

Block diagram of a basic feedback amplifier (single loop)

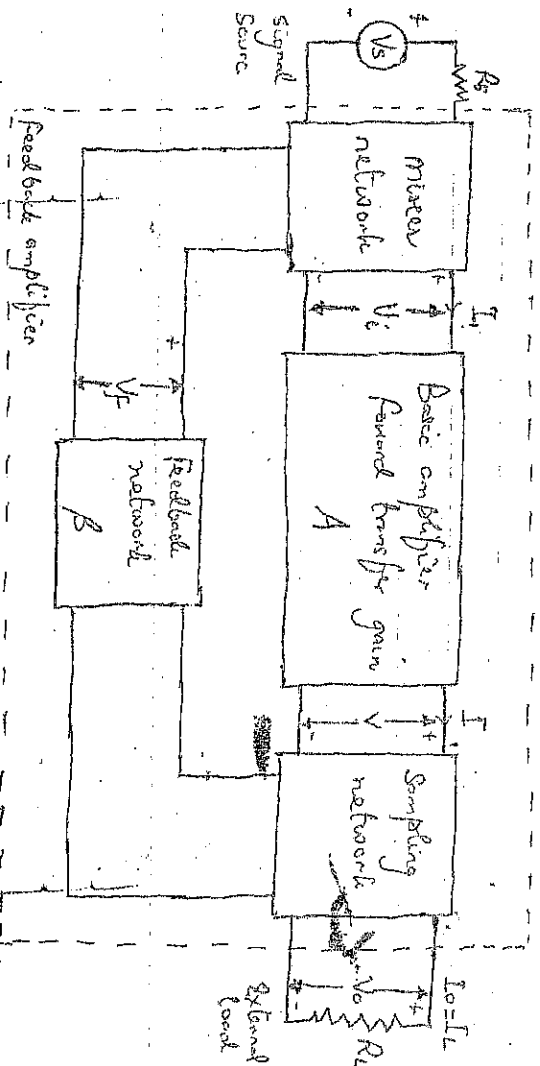


Fig 1.

The general diagram shows that the output voltage or current by means of a suitable sampling network and apply this signal to the input through a feedback two-port network.

At the input the feedback signal is combined with signal (source) signal through a mixer network and is fed into the amplifier proper. Also from above it can be seen that for negative feedback,

$$V_i = V_s - V_f \quad \text{--- difference signal ---} \quad (1)$$

$$V_f = V_o / \beta \quad \dots \quad (2)$$

(1)

and $v_o = A v_i$ ③
 Now from 1 and 2 and substituting in ③ yields

$$v_o = A(v_s - v_f)$$

$$v_o = A(v_s - v_o/\beta)$$

$$v_o (1 + A/\beta) = A v_s$$

$$v_o (1 + A/\beta) = A v_s$$

$$A v_f = \frac{v_o}{1 + A/\beta}$$

$$A v_f = \frac{A}{1 + A/\beta}$$

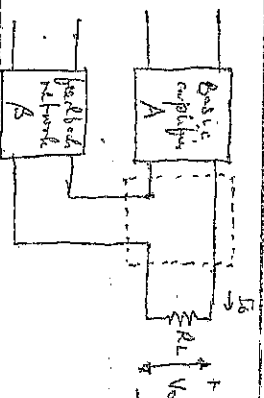
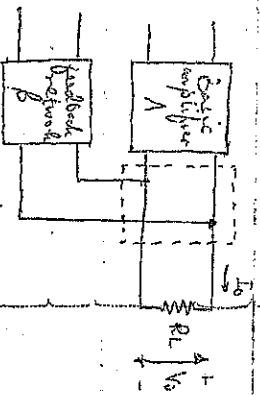
where A is the ^{open} loop gain
 and β is the feedback factor.

* Show that $A v_f = \frac{A}{1 + A/\beta}$ for positive feedback.

Note that in fig 1, the feedback network is usually a passive two-port network which may contain resistors, capacitors and inductors. Very often it is simply a resistive configuration.

NEGATIVE FEEDBACK TYPES

Shunt sampling blocks are shown in fig 2, called shunt and series.

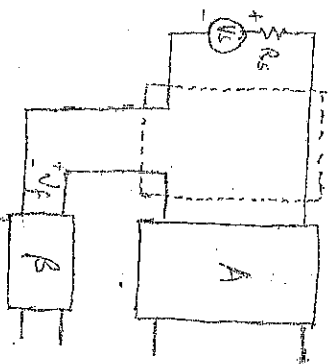


→ Voltage or shunt

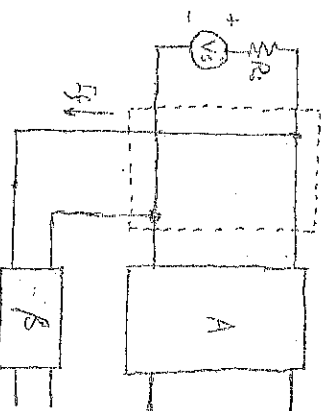
fig 2.

Current or series

Several mixing blocks are shown in fig 3, called series and shunt feedback.



Series feedback



Shunt feedback

fig 3

The above will combine to give the following configurations such as:

- Series-Current feedback
- Series-Voltage feedback
- Parallel(Shunt)-Current feedback
- Parallel(Shunt)-Voltage feedback

* Ask the students to draw all the configurations,

EFFECTS OF APPLYING NEGATIVE FEEDBACK TO AN AMPLIFIER

ON:-

✓ a) GAIN

The gain of an amplifier with feedback is reduced as compared with that of amplifier without feedback. But this has been compensated for by the following:

b) GAIN STABILITY

The gain of the amplifier will vary due to aging temperature replacement etc. To overcome this effect the amplifier must

✓ Prove

③

from feedback and can be shown from this formula that

$$A_{vf} = \frac{A}{1 + \beta A}$$

if the loop gain βA is made large enough by arrangement, $\beta A \gg 1$ and so

$$A_{vf} = \frac{A}{\beta A} = \frac{1}{\beta}$$

implying that the gain can be made to depend entirely on the feedback network and not on the active component. If the feedback network contains only stable passive elements, the improvement in stability may indeed be pronounced.

c) Distortion

The distortion could either be frequency, non-linear and delay distortion.

i) Frequency/delay distortion

is from $A_{vf} = \frac{1}{\beta}$, if the feedback network does not contain reactive elements, the overall gain is not a function of frequency. Under these circumstances a substantial reduction in frequency and phase shift (delay) distortion is obtained.

ii) Non-linear distortion

Suppose that a large signal is applied so that the operation of the active elements slightly beyond its range of linear operation and as a consequence the output signal is slightly distorted. Negative feedback is now introduced and the input signal is increased by the same amount by which the gain is reduced so that the output signal amplitude remains the same.

b) NOISE

It can be shown that the noise introduced in an amplifier is multiplied by the factor $\frac{1}{1 + \beta A}$ if feedback is employed. If $\beta A \gg 1$, this would seem to represent

Considerable reduction in output noise.

INPUT/OUTPUT IMPEDANCE

Negative feedback which samples the output voltage regardless of how this signal is returned to the input tends to decrease the output resistance. That which outputs the output current will tend to increase the output resistance.

If the feedback signal is returned to the input in series to oppose the applied voltage, regardless of whether it is obtained by sampling the output voltage or current, tends to increase the input resistance. That in which the output signal is fed back to the input is parallel tends to decrease the input resistance.

* Ask student to draw all the 8 possible combinations

F) BANDWIDTH / GAIN-BANDWIDTH PRODUCT

The bandwidth is related by the ω_c negative feedback whereas the gain-bandwidth product is not affected with or without feedback.

* demonstrate physically.

Evaluation

- 1) Assignment ✓
- 2) Test
- 3) Problems

Numerical Examples

Q1) An amplifier has an open-loop gain of 35. What is its voltage gain with 10% feedback? 1

From $A_{vf} = \frac{A}{1 + A\beta}$

$$A = 35$$

$$\beta = 0.1$$

$$\therefore A_{vf} = \frac{35}{1 + 35 \times 0.1} = \frac{35}{4.75} = \frac{35}{4.8} = 7.7$$

from approximate formula when $A\beta \gg 1$

$$A_{vf} = \frac{1}{\beta} = \frac{1}{0.1} = 10$$

TUTORIALS

1. When voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50. Calculate the fraction of the output voltage feedback. If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75

2. An amplifier having a gain of 500 with feedback has negative feedback applied which reduces the overall gain to 100. Calculate the fraction of the output voltage feedback. If due to aging of components, the gain without feedback falls by 20%, calculate the percentage fall in gain with feedback.

3. An amplifier having a gain of 100 having 10% voltage negative feedback applied in series with the input signal. Calculate the overall stage gain with feedback. If supply voltage varies causing the gain with feedback to fall by 10%, determine the percentage change in gain without feedback.

(6)

4) The gain and distortion of an amplifier are 150 and 5% respectively without feedback. If the stage has 10% of its output voltage applied as negative feedback, find the distortion of the amplifier with feedback.

5) An amplifier has a current gain of 240 and input impedance of 15 k Ω without feedback. If negative current feedback of -0.015 is applied, what will be the input impedance of the amplifier?

6) An amplifier has a current gain of 250 and a bandwidth of 40 kHz without feedback. If negative current feedback of 1% is applied, what is the bandwidth of the amplifier.

7) An amplifier with current gain of 250 and output impedance of 3 k Ω without feedback. If negative current feedback of 1% is applied, what will be the output impedance of the amplifier?

7

① Δ_{cur}

$$A = 100$$

$$A_y = 50$$

$$k = ?$$

Soln

$$A_y = \frac{A}{1 + k}$$

$$50 = \frac{100}{1 + 100k}$$

$$100 = 50(1 + 100k)$$

$$100 = 50 + 5000k$$

$$100 - 50 = 5000k$$

$$k = \frac{50}{5000} = 0.01$$

Soln

2% -

$$k = 0.02$$

$$A_y = 75$$

$$75 = \frac{A}{1 + 0.02k}$$

$$A = 75(1 + 0.02k)$$

$$A = 75 + 0.02kA$$

$$A - 0.02kA = 75$$

$$(1 - 0.02k)A = 75$$

$$0.98A = 75$$

$$A = \frac{75}{0.98} \approx 76.53$$

② Δ_{cur}

$$A = 500$$

$$A_y = 100$$

$$k = ?$$

Soln

$$A_y = \frac{A}{1 + k} \Rightarrow 100 = \frac{500}{1 + 500k}$$

$$100(1 + 500k) = 500$$

$$100 + 50000k = 500$$

$$k = \frac{500 - 100}{50000} = 0.008$$

Soln

2%

$$2\% \text{ fall of } A = \frac{80}{100} \times 500 = 400$$

$$A = 400$$

$$A_y = \frac{A}{1 + k} = \frac{400}{1 + 400 \times 0.008}$$

$$A_y = 95.24$$

$$\% \text{ fall in } A_y = \frac{100 - 95.24}{100} \times 100$$

$$= 4.76\%$$

①

③ Ans

$$A = 150$$

$$B = 9\% = \frac{9}{100} = 0.09$$

Sol

$$A_f = \frac{A}{1 + B} = \frac{150}{1 + 150 \times 0.09}$$

$$A_f = \underline{10}$$

$$15\% \text{ fall in } A_f = \frac{90}{150} \times 100 = 9$$

$$A_{ef} = \frac{A}{1 + B} \Rightarrow 9 = \frac{A}{1 + 150 \times 0.09}$$

$$A = 9(1 + 0.09A)$$

$$A = 9 + 0.81A$$

$$A - 0.81A = 9$$

$$A(1 - 0.81) = 9$$

$$0.19A = 9$$

$$A = \frac{9}{0.19} = 47.4$$

$$\% \Delta A = \frac{A_{old} - A_{new}}{A_{old}} \times 100\%$$

$$= \frac{150 - 47.4}{150} \times 100\%$$

$$= \underline{68.6\%}$$

④ Ans

$$A = 150, D = 5\% = \frac{5}{100} = 0.05$$

$$B = 18\% = \frac{18}{100} = 0.18, D_f = ?$$

Sol

$$D_f = \frac{D}{1 + B}$$

$$= \frac{0.05}{1 + 150 \times 0.18} = \frac{0.05}{10}$$

$$= 0.00313 \times 100\%$$

$$= \underline{0.313\%}$$

⑤ Ans

$$A = 240, Z_1 = 15k\Omega$$

$$B = 0.615, Z_{if} = ?$$

$$Z_{if} = \frac{Z_1}{1 + B}$$

$$= \frac{15}{1 + 240 \times 0.615} = \frac{15}{4.8}$$

$$Z_{if} = \underline{3.125k\Omega}$$

⑥

②

⑥ Soln

$$A = 250, B_W = 450 \text{ kHz}, \beta = 1\% = \frac{1}{100} = 0.01$$

Soln

$$B_{WF} = B_W(1 + \beta)$$

$$B_{WF} = 450(1 + 250 \times 0.01)$$

$$B_{WF} = \underline{\underline{1450 \text{ kHz}}}$$

⑦ Soln

$$A = 200, Z_o = 8 \text{ k}\Omega, \beta = 1\% = \frac{1}{100} = 0.01$$

Soln

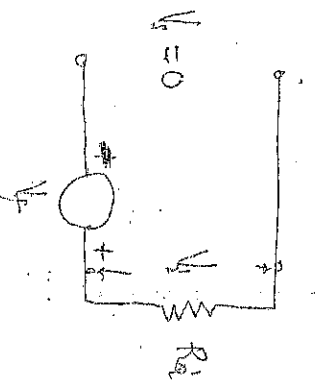
$$Z_{of} = Z_o(1 + \beta)$$

$$Z_{of} = 8(1 + 200 \times 0.01)$$

$$Z_{of} = \underline{\underline{9 \text{ k}\Omega}}$$

⑧

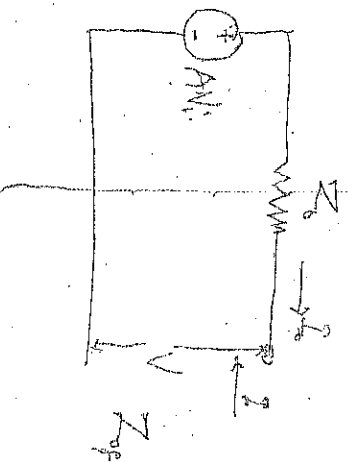
$$V_o = A V_i$$



$$I = \frac{V - A V_i}{Z_o}$$

$$\begin{aligned} \text{Since } V_i &= V_o - V_f \\ V_o &= -V_f \end{aligned}$$

$$V_f = \beta V \text{ or } -\beta V$$

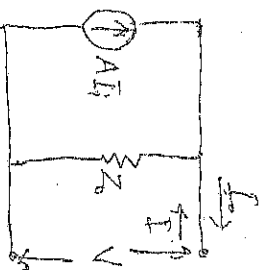
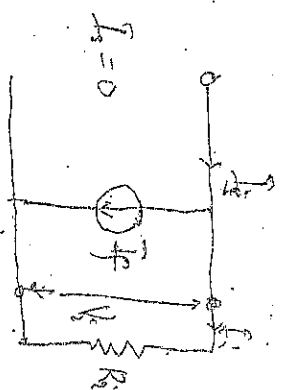


$$I = \frac{V - A(-\beta V)}{Z_o}$$

$$I = \frac{V + A\beta V}{Z_o}$$

$$I = \frac{V(1 + A\beta)}{Z_o}$$

$$\frac{V}{I} = \frac{Z_o}{1 + A\beta}$$



$$I = \frac{V}{Z_o} - A I_i$$

$$\begin{aligned} \text{Since } I_o &= I - I_f \\ I_o &= -I_f \end{aligned}$$

$$I_f = \beta I_o$$

$$I_o = -I$$

$$I = \frac{V}{Z_o} - A\beta I$$

$$\begin{aligned} I + A\beta I &= \frac{V}{Z_o} \\ I(1 + A\beta) &= \frac{V}{Z_o} \end{aligned}$$

$$\begin{aligned} I_f &= \beta I \\ \therefore I_o &= \beta I \end{aligned}$$

$$\frac{V}{I} = Z_o(1 + A\beta)$$